

Phase Manifold Model: First Principles and Mathematics

Time is angular, space is generated in realization, and depth is different

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Version 3.7.1 · September 30, 2026
Core Revision v62

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Abstract

The Phase Manifold Model (PMM) describes physical reality through complete events organized as Before, During, and After. B = Before, D = During, and A = After.

One complete event is a Field Transition Unit (FTU):

$$F_{tu} = B \circ D \circ A$$

PMM treats time as angular, space as emergent during D, and depth as a relational D-phase coordinate distinct from ordinary height and width. The FTU moves through four c-order grades, from c^0 to c^3 . The middle grades belong to realized space during D, while c^3 marks the transition into the post-spatial After phase. During D, the realized geometry grows and the two c-grade vantages move toward agreement. Contact @ opens the realized phase, and anti-contact -@ closes it. A retains the completed event as a persistent record. Across many FTUs, those records accumulate in Φ and contribute to gravity. Action gives B, D, and A a common measure.

PMM IN ONE PAGE

PMM begins with a physical event: possibility becomes realized interaction, and completed interaction leaves a persistent record.

$$B \rightarrow D \rightarrow A$$

Before (B) carries pre-spatial possibility. **During (D)** carries active realization and spatial geometry. **After (A)** carries the completed post-spatial record. PMM's human-facing correspondences are B = Wave, D = Particle, and A = Cube. One complete event is a Field Transition Unit:

$$F_{tu} = B \circ D \circ A$$

The FTU physical c-order representation is:

$$B = (c^0)_{mp}$$

$$D = (c^1, c^2)c$$

$$A = (c^3)k_B$$

With its complete representation as:

$$F_{tu}^c = ((c^0)_{mp}, (c^1, c^2)c, (c^3)k_B)$$

Its c-order progression is:

$$c^0 \rightarrow c^1 \rightarrow c^2 \rightarrow c^3$$

The raw physical magnitudes are:

$$c^0 = 1$$

$$c^1 = 299,792,458 \text{ m/s} = d/t$$

$$c^2 = 89,875,517,873,681,764 \text{ m}^2/\text{s}^2 = E/m \text{ (massive sector)} = E^2/|\mathbf{p}|^2 \text{ (null sector)}$$

$$c^3 = 26,944,002,417,373,989,539,335,912 \text{ m}^3/\text{s}^3 = \hbar G_N / \ell_p^2$$

The numerical identities are standard physics; PMM's physical claim is their ordered assignment to the B→D→A event grades.

Each adjacent grade is generated by one additional multiplicative factor of c. Within PMM, B is assigned c^0 , D is assigned c^1 and c^2 , and A is assigned c^3 . The c^1/c^2 pair supplies the D-side physical structure from which relational spatial geometry and downstream GR recovery are built.

The angular c-order profile is the grade-by-grade angular/action readout across $c^0 \rightarrow c^1 \rightarrow c^2 \rightarrow c^3$. It describes the internal architecture of a single FTU; it does not imply that macroscopic motion or spacetime must appear piecewise-linear.

During D, two derived readouts track the same angular progression:

$$\text{realized scale expansion: } L = \ell_E \theta^D$$

$$\text{vantage-imbalance contraction: } r = r_0 \cos(\theta^D/2)$$

A normalized pair is:

$$X^D = q = \theta^D/\pi$$

$$C^D = \Theta_{\text{out}} = \cos(\theta^D/2)$$

so X^D runs $0 \rightarrow 1$ while C^D runs $1 \rightarrow 0$ across D.

Contact and Realization

The event sequence is:

$$B \rightarrow @ \rightarrow D \rightarrow -@ \rightarrow A$$

@ denotes contact actualization: an admissible B possibility enters realized D. -@ denotes the terminal D boundary. D contains active xyz spatial realization; depth z is active only in D. At the $D \rightarrow A$ transition, active spatial depth ends and the completed event is transcribed into the persistent A record.

C_1 , C_2 , and C_3 organize entry, realized path, and persistence/transcription. Θ_{con} is the contact-information structure governing the conditions around @; it is not an alternate symbol for the @ event itself:

$$\Theta_{\text{con}} = \{C_1[@,\Phi], C_2[\Phi,\delta,\Sigma], C_3[\Phi,\Sigma]\}$$

PMM uses two complementary vantages:

Single-FTU vantage = internal $B \rightarrow D \rightarrow A$ event geometry.

Multi-FTU vantage = relational geometry among FTUs.

An FTU has no isolated physical realization; physical geometry is relational among FTUs.

PART I — CORE ARCHITECTURE

Compact First Principles

This section collects the core PMM definitions, representations, and physical readouts in one compact reference.

Event hierarchy

$T = \text{Time}$

$B = \text{Before} = \text{Potential} = \text{Quantum Mechanics} = \text{Wave} = \text{Uncertainty} = \text{Pre-spatial}$

$D = \text{During} = \text{Realization} = \text{General Relativity} = \text{Particle} = \text{Partial Certainty} = \text{Spatial xyz}$

$A = \text{After} = \text{Transcription} = \text{Information Theory} = \text{Cube} = \text{Persistent Record} = \text{Post-spatial}$

$T \approx B \circ D \circ A$

$@ = \text{Wave Function Collapse / Contact}$

$-@ = \text{Anti Contact}$

Spatial geometry and contact window

$x = \text{Height}$

$y = \text{Width}$

$z = \text{Depth}$

$\Theta_{\text{in}} = \text{opening factor} — \text{spatial manifestation grows from } @ \text{ through } D; \text{ legacy name: Birth Sigmoid}$

$\Theta_{\text{out}} = \text{closing factor} — \text{spatial manifestation closes toward } -@; \text{ legacy name: Record Sigmoid}$

$\text{Space Window} = \text{the bounded region where space exists between the opening and the closing}$

$\Theta_{\text{win}} = \text{Space Window} = \text{region where geometry is active}$

Constants, fields, and constraints

k_B = Boltzmann Constant

c = Speed of Light Constant

m_p = Planck Mass

Φ = Accumulated record field

δ = Local interaction depth

Σ = Closure spread

d = realized D traversal length used in $T^D = d/c$

t = coordinate time used in $c^1 = d/t$

τ = proper time; $d\tau^D = D$ proper-time increment

C_1 = Entry Constraint $\rightarrow$ What can enter the channel

C_2 = Path Constraint $\rightarrow$ What conditions the interaction

C_3 = Exit Constraint $\rightarrow$ What persists and is transcribed into Φ

Θ_{con} = Contact-information structure governing the conditions around @

$$\Theta_{\text{con}} = \{C_1[@, \Phi], C_2[\Phi, \delta, \Sigma], C_3[\Phi, \Sigma]\}$$

FTU architecture

$$F_{\text{tu}} = B \circ D \circ A$$

The FTU physical c-order representation is:

$$B = (c^0)_{\text{mp}}$$

$$D = (c^1, c^2)c$$

$$A = (c^3)k_B$$

With its complete representation as:

$$F_{\text{tu}}^c = ((c^0)_{\text{mp}}, (c^1, c^2)c, (c^3)k_B)$$

$$c^1 \mapsto (x_1^D, y_1^D) = \text{first D spatial vantage}$$

$$c^2 \mapsto (x_2^D, y_2^D) = \text{second D spatial vantage}$$

$$z^D = \text{relational depth generated across the two D vantages}$$

FTU family glossary

$$F_{tu} = \text{complete } B \rightarrow D \rightarrow A \text{ event}$$

$$F_{tu}^c = \text{c-order representation}$$

$$F_{tu}^M = \text{mass readout}$$

$$F_{tu}^E = \text{energy readout}$$

$$F_{tu}^T = \text{time readout}$$

$$F_{tu}^{xy} = \text{spatial readout}$$

$$F_{tu}^G = \text{gravity readout}$$

$$F_{tu}^s = \text{action readout}$$

$$F_{tu}^{sv} = \text{phase-action vector}$$

$$\text{Mass} = F_{tu}^M = (m_p, m, \hbar\omega^A/c^2)$$

$$\text{Energy} = F_{tu}^E = (m_p c^2, mc^2, \hbar\omega^A)$$

$$\text{Time} = F_{tu}^T = ([\hbar/(m_p c^2)]\Delta\theta^B, d/c, \Delta\theta^A/\omega^A)$$

$$\text{Spatial} = F_{tu}^{xy} = (\psi(x,y,z), xyz, xy)$$

$$\text{Gravity} = F_{tu}^G = \nabla\varphi; g = c^2 F_{tu}^G$$

$$\text{Action} = F_{tu}^s = S^B + S^D + S^A$$

Physical c-order profile

$$c^0 = 1$$

$$c^1 = 299,792,458 \text{ m/s} = d/t$$

$$c^2 = 89,875,517,873,681,764 \text{ m}^2/\text{s}^2 = E/m \text{ (massive sector)} = E^2/|\mathbf{p}|^2 \text{ (null sector)}$$

$$c^3 = 26,944,002,417,373,989,539,335,912 \text{ m}^3/\text{s}^3 = \hbar G_N/\ell_p^2$$

The numerical identities are standard physics; PMM's physical claim is their ordered assignment to the $B \rightarrow D \rightarrow A$ event grades.

D expansion / contraction readout:

$$X^D = q = \theta^D / \pi : 0 \rightarrow 1$$

$$C^D = \theta_{\text{out}} = \cos(\theta^D / 2) : 1 \rightarrow 0$$

The c-order ladder is the event-grade architecture of the FTU. The numerical escalation comes directly from exponentiation.

$$c^0 \leftrightarrow m_p$$

$$c^1 \leftrightarrow c$$

$$c^2 \leftrightarrow c$$

$$c^3 \leftrightarrow A \text{ transcription grade}$$

The crease roles are distinct:

$$m_p = \text{entry crease}$$

$$c = \text{realization crease}$$

$$k_B = \text{transcription crease}$$

k_B is not an additional c-order scaling step. D closes after the c^2 realization grades; c^3 is the A-side transcription grade that carries the completed event into persistent record.

Core physical interpretations

Orientation: B-facing outwardness $\rightarrow$ A-facing inwardness

Gravity: gradient in D-phase time behavior

Light: "still" in the zero-proper-time sense

Entropy: A-phase persistent-record quantity

Space: emergent during D

Depth: distinct from ordinary height/width

Time: angular

FTU: full $B \rightarrow D \rightarrow A$ event

FTU is geometry

Single-FTU vantage = Internal $B \rightarrow D \rightarrow A$ event geometry

Multi-FTU vantage = Relational geometry among FTUs

An FTU has no isolated physical realization; physical geometry is relational among FTUs

Wave / Particle / Cube: B / D / A

Constants: phase creases

Depth is active in D and ends at transcription. After is accessed through retained information expressed in x/y representation.

Action provides a common measure across B, D, and A.

Two Vantages of PMM

PMM separates its architecture into two complementary vantages:

Single FTU: creases + c-order + action

Multi-FTU: Φ + timing response + gravity

The single-FTU vantage describes the internal structure of one complete $B \rightarrow D \rightarrow A$ event. Its constants act as creases, the Ladder of Light supplies the c-order grade structure, and action provides the common measure across the phases.

The multi-FTU vantage describes how completed events influence later events. Persistent record accumulates in Φ , Φ conditions later D-phase timing, and relational differences in that timing produce the gravity response.

The two vantages are connected by completed record, but they do not require the same internal machinery. The c-order ladder defines the architecture of a single FTU; gravity is a relational multi-FTU phenomenon.

PART II — SINGLE-FTU VANTAGE

1. Fundamental Event

The Field Transition Unit is:

$$F_{tu} = B \circ D \circ A$$

with the phase boundaries:

$$B \rightarrow @ \rightarrow D \rightarrow -@ \rightarrow A$$

where:

- @ = contact / wave-function collapse / entry into realized D
- -@ = anti-contact / terminal D boundary / exit from realized D into A

The single-FTU vantage describes the internal geometry of one complete event.

The multi-FTU vantage describes relational geometry among FTUs.

An FTU has no isolated physical realization. Physical world geometry is relational among FTUs.

PMM 3.7.1 develops the single-FTU vantage in Part II and the relational multi-FTU layer in Part III.

2. c-Order Event Ladder

The FTU uses four event grades:

$$\begin{array}{ccccc} B & & D & & A \\ c^0 & & c^1 & c^2 & c^3 \end{array}$$

Interpretation:

$$c^0 = \text{pre-spatial grade}$$

$$c^1 = \text{first D realization grade / first x-y vantage}$$

$$c^2 = \text{second D realization grade / second x-y vantage}$$

$$c^3 = \text{post-spatial transcription grade}$$

PMM 3.7.1 uses the core D identification:

$$c^1 \mapsto V_1 = (x_1^D, y_1^D)$$

$$c^2 \mapsto V_2 = (x_2^D, y_2^D)$$

The arrow $\mapsto$ denotes a physical readout correspondence rather than dimensional equality. c^1 and c^2 are realization grades; each supplies an x-y vantage on the same local D geometry. The c-order grades are therefore not themselves spatial axes.

The event-grade assignment is:

$$B = (c^0)_{mp}$$

$$D = (c^1, c^2)c$$

$$A = (c^3)k_B$$

With its complete representation as:

$$F_{tu}^c = ((c^0)_{mp}, (c^1, c^2)c, (c^3)k_B)$$

The native FTU ladder terminates at c^3 because the complete event contains:

1 B grade + 2 D grades + 1 A grade

The native FTU event grades are c^0 through c^3 .

Physical c-Order Architecture — The Ladder of Light

PMM postulates the c-order ladder as the **physical event-grade architecture of the FTU**:

$$c^0 \rightarrow c^1 \rightarrow c^2 \rightarrow c^3$$

The numerical escalation across c^0 , c^1 , c^2 , and c^3 comes directly from exponentiation. The Ladder of Light is the physical claim that the $B \rightarrow D \rightarrow A$ event architecture occupies this ordered structure in the correspondence:

$$F_{tu}^c = ((c^0)_{mp}, (c^1, c^2)c, (c^3)k_B)$$

The symbols in this tuple do not all represent the same kind of multiplication. They encode three crease roles:

m_p = entry crease

c = realization crease

k_B = transcription crease

The realization sequence is:

$$B / \text{entry} \rightarrow c^0$$

$$D_1 / \text{realization} \rightarrow c^1$$

$$D_2 / \text{realization} \rightarrow c^2$$

$$A / \text{transcription} \rightarrow c^3$$

The repeated c operation scales the event through the ordered grades. D occupies c^1 and c^2 . At $-@$, D closes and the event enters c^3 , the A-side transcription grade.

A is therefore the final grade of the native c -order architecture, not a c^4 extension. k_B marks the transcription of the completed event into persistent record; it does not create another c -order grade.

This distinction is central:

c -order = event-grade architecture

k_B = transcription into a different category of quantity

action = common ruler across B , D , and A

The c^1 and c^2 grades supply two realized D vantages:

$$c^1 \mapsto V_1 = (x_1^D, y_1^D)$$

$$c^2 \mapsto V_2 = (x_2^D, y_2^D)$$

Their relation supplies the D -side grade geometry. The variable r measures imbalance between the two vantage scales; relational depth and closure are built from their common geometry.

Logical status: the c -order correspondence is a PMM first-principle postulate for the single-FTU architecture. The exponentiation supplies the numerical scale; PMM supplies the physical assignment of those grades to entry, realization, D closure at $-@$, and A-side transcription.

The c -Order / Crease Map

The Ladder of Light is best drawn as a **grade-and-crease map**, not as one numerical curve containing B , D , and A magnitudes.

Along the c -order axis:

$$c^0 \rightarrow c^1 \rightarrow c^2 \rightarrow c^3$$

The crease roles are:

$$m_p \rightarrow \text{entry}$$

$$c \rightarrow \text{realization across the D grades}$$

$$k_B \rightarrow \text{transcription at the terminal D} \rightarrow \text{A boundary}$$

The event-grade climb therefore ends at c^3 , the A transcription grade.

The D $\rightarrow$ A transition is:

$$c^2 \text{ completion} \rightarrow -@ \rightarrow c^3 / \text{A transcription} \rightarrow \text{persistent record}$$

The change at $-@$ is a transition from realized spatial geometry to persistent record, not a new c^4 grade.

c^3 Planck-Scale Remark and Gravitational-Crease Candidate

Standard-physics remark. At the terminal c-order grade, the Planck-length definition may be written:

$$\ell_p = \sqrt{(\hbar G_N / c^3)}$$

Equivalently:

$$c^3 = \hbar G_N / \ell_p^2$$

This is the standard Planck-length identity written in reverse. PMM assigns this standard relation to the terminal c^3 / A grade as an architectural correspondence.

PMM closure-area readout. The current two-vantage geometry gives:

$$x^D y^D = L^2$$

and at completed D closure:

$$L_{cl} = \pi \hbar c / E_{cl}^D$$

Define the completed planar closure-area readout:

$$A_{cl} = x_{cl}^D y_{cl}^D = L_{cl}^2$$

therefore:

$$A_{cl} = (\pi \hbar c / E_{cl}^D)^2$$

Define the gravitational-crease area from the same standard Planck relation:

$$A_G = \hbar G_N / c^3 = \ell_p^2$$

PMM then isolates a narrower candidate condition for the gravitational-crease limiting case. In ladder language, this is the condition that completed D closure reaches the Planck-area scale associated with the terminal c^3 correspondence:

$$A_{cl} = A_G$$

If native C_3 persistence dynamics select this equality at that limiting boundary, then:

$$(\pi \hbar c / E_{cl}^D)^2 = \hbar G_N / c^3$$

which gives:

$$E_{cl}^D = \pi \sqrt{(\hbar c^5 / G_N)} = \pi m_p c^2$$

The factor π comes directly from the current closure postulate $\theta_{cl} = \pi$ through $L_{cl} = \pi \hbar c / E^D$.

and consequently:

$$L_{cl} = \ell_p$$

$$A_{cl} = \ell_p^2$$

The equality $A_{cl} = A_G$ is a candidate gravitational-crease condition rather than a universal closure rule. Ordinary completed FTUs retain carrier-dependent L_{cl} and A_{cl} . This candidate belongs to the single-FTU closure/transcription sector and does not enter the multi-FTU gravity chain $\Phi \rightarrow \kappa \rightarrow \varphi \rightarrow g$ as a direct dynamical input.

The terminal architectural sequence remains:

$$c^2 \rightarrow -@ \rightarrow c^3 / k_B \text{ transcription} \rightarrow A \text{ record}$$

k_B remains the transcription crease beside terminal c^3 ; it is separate from the Planck identity and the closure-area candidate.

Action as the Ruler Across the Architecture

The c-order grades carry different dimensional character, and c^3 is the A-side transcription grade rather than another D spatial grade. PMM therefore uses action as the common measure across the B/D/A event:

$$F_{tu}^{sv} = (S^B, S^D, S^A)$$

with:

$$[S^B] = [S^D] = [S^A] = \text{action}$$

The additive action ledger is:

$$F_{tu}^s = S^B + S^D + S^A$$

Action provides the common ruler across the phase-distinct B, D, and A sectors.

The graphical consequence is simple: the c-order plot rises through B and D to c^2 , then reaches c^3 at A transcription. The D→A boundary changes the physical category from realized space to persistent record without adding a c^4 grade.

3. Creases

The phase creases are:

$$B \leftrightarrow m_p$$

$$D \leftrightarrow c$$

$$A \leftrightarrow k_B$$

Interpretation:

- m_p = entry crease
- c = realization crease
- k_B = transcription crease

Technical role:

$$m_p \rightarrow \text{anchors entry into the FTU architecture}$$

$$c \rightarrow \text{scales D progression through the realization grades}$$

$$k_B \rightarrow \text{transcribes A-side record multiplicity into entropy units}$$

The crease roles are not identical operations. c is the realization-scaling crease; k_B is the transcription crease and does not extend the c-order ladder beyond c^3 .

4. Single-FTU Readouts

The primary FTU readouts are:

$$\begin{aligned} \text{Mass} &= F_{tu}^M = (m_p, m, \hbar\omega^A/c^2) \\ \text{Energy} &= F_{tu}^E = (m_p c^2, mc^2, \hbar\omega^A) \\ \text{Time} &= F_{tu}^T = ([\hbar/(m_p c^2)]\Delta\theta^B, d/c, \Delta\theta^A/\omega^A) \end{aligned}$$

Interpretation:

$$B \text{ time} = \text{projective potential span over the intrinsic conversion scale}$$

$$D \text{ time} = \text{realized traversal time}$$

$$A \text{ time} = \text{record span over the intrinsic transcription scale}$$

The compact mathematics below expresses these readouts in the PMM architecture.

4.1 Action as the Common B/D/A Measure

Action provides a common measure across B, D, and A. The c-order ladder assigns c^0 to B, c^1, c^2 to D, and c^3 to A transcription. Action provides the commensurable ruler across the full event.

Write the phase-action representation as:

$$F_{tu}^{sv} = (S^B, S^D, S^A)$$

and the additive FTU action ledger as:

$$F_{tu}^s = S^B + S^D + S^A$$

where S^B , S^D , and S^A are the action contributions associated with the three sequential phases. The ledger records those contributions in one common physical measure. It does not drive realization or transcription.

PMM uses action as the common B/D/A ruler but does **not** impose the fixed bookend normalization $S^B=S^A=\hbar$. Such a normalization would force unit spans by definition. The current construction instead uses phase-specific dimensionless spans:

$$S^B = \hbar \Delta\theta^B$$

$$S^A = \hbar \Delta\theta^A$$

and for D angular progression:

$$dS^D = \hbar d\theta^D$$

therefore:

$$S^D(\theta^D) = \hbar \theta^D$$

with $S^D(0)=0$. Since the current single-FTU closure postulate is:

$$\theta_{cl} = \pi$$

the completed D action readout is:

$$S^{D,cl} = \pi\hbar$$

The candidate B span is tied to the physical C_1 -selected contact channel. For the actualized channel $j@$ with contact weight $p_{j@}$, define the projective contact span:

$$\Delta\theta^B = \arccos\sqrt{p_{j@}}$$

For a normalized pure B state and rank-one contact channel this is the ray-space angle between the B state and the selected contact channel. It is invariant under global phase, remains finite at the endpoints, and inherits relative-phase dependence through the C_1 projection whenever the physical contact basis differs from the representation basis.

The A span is tied to the information carried by the actualized C_1 -selected contact channel. For the realized channel $j@$, define the single-event A information readout:

$$I^A = -\ln p_{j@}$$

and the associated effective A-side multiplicity:

$$\Omega_A = e^{I^A} = 1/p_{j@}$$

Therefore:

$$\Delta\theta^A = \mathcal{S}^A/k_B = \ln \Omega_A = I^A = -\ln p_{j@}$$

with $\mathcal{S}^A = k_B I^A$. For N equiprobable alternatives, $p_{j@} = 1/N$ and $\Omega_A = N$. The A span is an information span; it is not the existence time of A.

The completed elementary FTU action ledger is therefore:

$$F_{tu}^{s,cl} = \hbar(\Delta\theta^B + \pi + \Delta\theta^A)$$

The B span rule is a **candidate phase-action readout**; the A span rule is an **information/action readout**. They standardize the phases in action units without claiming that the c-order values themselves are actions or that the span rules cause contact, closure, or transcription.

The D action is carrier-general because it is defined from angular progression. In the null sector, $d\tau^D=0$ while $d\theta^D \neq 0$, so null D retains a defined angular-action readout without identifying action with proper time.

4.2 c-Order Readout in the Common Action Unit

The c-order grades and action are distinct quantities. The common phase-action ruler maps the grades into action readouts:

c-order grade	PMM role	Action readout	Normalized $S/\hbar$	UOM
c^0	B	$S^B = \hbar\Delta\theta^B$	$\Delta\theta^B$	J·s
c^1	D vantage 1	$S^D = \hbar\theta^D$; at closure $\pi\hbar$	θ^D ; at closure π	J·s
c^2	D vantage 2	$S^D = \hbar\theta^D$; at closure $\pi\hbar$	θ^D ; at closure π	J·s
c^3	A	$S^A = \hbar\Delta\theta^A$	$\Delta\theta^A$	J·s

The c^1 and c^2 rows are two vantages of the same D phase. They therefore share the same S^D readout and are not counted as two separate D action contributions in the additive FTU ledger.

At completed single-FTU closure, the c-order action-readout profile is therefore:

$$\Delta\theta^B \rightarrow \pi \rightarrow \pi \rightarrow \Delta\theta^A$$

in units of $\hbar$. The B and A values are event-dependent rather than universal constants. This profile is a common-unit readout of the c-order architecture, not a replacement for the raw $c^0 \rightarrow c^1 \rightarrow c^2 \rightarrow c^3$ event-grade values.

5. B Phase — Potential

B is the pre-spatial possibility phase.

A B state is represented by:

$$\Psi^B = \sum_i a_i e^{i\theta_i} |i\rangle$$

with:

$$\sum_i |a_i|^2 = 1$$

C_1 controls entry into D.

For a two-component representation of the B state:

$$p_+ = |a_+|^2$$

$$p_- = |a_-|^2$$

$$p_+ + p_- = 1$$

These representation weights need not themselves be the physical contact weights unless the representation basis is already the C_1 -selected contact basis.

5.1 Candidate C_1 Contact-Channel Selection

PMM tests the following minimal completion of the entry constraint. Let $C_1(\Phi)$ select the admissible physical contact channels represented by projectors $\{\Pi_j\}$:

$$C_1(\Phi) \rightarrow \{\Pi_j\}$$

For B state Ψ^B , define the normalized contact weights:

$$p_j@ = \langle \Psi^B | \Pi_j | \Psi^B \rangle / \sum_k \langle \Psi^B | \Pi_k | \Psi^B \rangle$$

For the two contact channels that seed the two D vantages:

$$p_1@ + p_2@ = 1$$

In the nondegenerate two-channel case, $j@ \in \{1,2\}$ and $0 < p_j@ < 1$. This keeps the ratio coordinate r_0 finite and identifies $p_j@$ as the actualized-channel weight used by $\Delta\theta^B$ and I^A .

PMM then adopts the **B→D ratio-transfer postulate** at contact:

$$\lim_{\theta^B \rightarrow 0^+} \rho_1/\rho_2 = p_1@ / p_2@$$

Equivalently:

$$e^{2r_0} = p_1@ / p_2@$$

Since the D-vantage scale ratio is defined by:

$$\rho_1 / \rho_2 = e^{2r}$$

the initial D relational coordinate follows algebraically:

$$r_0 = \frac{1}{2} \ln(p_1@/p_2@)$$

Relative phase in Ψ^B can contribute to the contact weights through the projections whenever the C_1 -selected contact basis differs from the representation used to write Ψ^B . Relative phase therefore need not be introduced as an additional D coordinate.

Thus:

$$\begin{aligned} & \text{full B state } \Psi^B \\ & \rightarrow C_1\text{-selected contact channels} \\ & \rightarrow \text{contact weights } p_j@ \\ & \rightarrow \text{D-vantage imbalance} \end{aligned}$$

C_1 can therefore be read intuitively as:

what enters the channel

Status: candidate contact-channel selection rule. It is not yet derived from deeper PMM structure.

5.2 Candidate B Projective Span

Once C_1 physically selects the contact channels, the selected channel supplies a non-arbitrary endpoint for a B-side projective span. For the actualized channel $j@$:

$$\Delta\theta^B = \arccos\sqrt{p_j@}$$

with:

$$p_j@ = \langle \Psi^B | \Pi_j | \Psi^B \rangle / \sum_k \langle \Psi^B | \Pi_k | \Psi^B \rangle$$

The associated B phase time and action readouts are:

$$T^B = [\hbar/E^B] \Delta\theta^B$$

$$S^B = E^B T^B = \hbar \Delta\theta^B$$

For the principal B energy $E^B = m_p c^2$:

$$T^B = [\hbar/(m_p c^2)] \Delta\theta^B$$

The intrinsic unit-span conversion scale remains $T^{B*} = \hbar/E^B$; the physical B phase readout need not equal that unit span.

Status: candidate projective B span. The contact basis is physically supplied by C_1 rather than chosen as an arbitrary reference.

5.3 Candidate Reciprocal Closure Pairing

PMM tests a reciprocal B-side closure pairing as a candidate route to the Born rule. For a finite or countable C_1 -selected contact/closure basis, the candidate probability readout is:

$$p_j = |a_j|^2 / \sum_k |a_k|^2$$

When the channels are represented by orthogonal projectors, this is the component form of the contact-weight rule above. The candidate is tested for positivity, additivity, normalization, reciprocal symmetry, and closure-basis invariance. It remains a candidate unless those tests hold and the relation follows from PMM-native structure.

6. D Phase — Realization

D is the spatial realization phase.

The two D realization grades provide two local spatial vantages:

$$c^1 \mapsto V_1 = (x_1^D(\theta^D), y_1^D(\theta^D))$$

$$c^2 \mapsto V_2 = (x_2^D(\theta^D), y_2^D(\theta^D))$$

These are two vantages on one local D geometry rather than four independent spatial dimensions.

Define the scalar scale of each vantage:

$$\rho_1 = \sqrt{(x_1^D y_1^D)}$$

$$\rho_2 = \sqrt{(x_2^D y_2^D)}$$

Define the common D scale and the c-grade imbalance coordinate:

$$L = \sqrt{(\rho_1 \rho_2)}$$

$$r = \frac{1}{2} \ln(\rho_1 / \rho_2)$$

These invert exactly:

$$\rho_1 = L e^r$$

$$\rho_2 = L e^{-r}$$

Define the D energy length:

$$\ell_E = \hbar c / E$$

PMM adopts the **D scale postulate**:

$$L = \ell_E \theta^D$$

PMM separately adopts the minimal harmonic closure implementation:

$$r'' + \frac{1}{4}r = 0$$

Equivalently:

$$d^2/d\theta^{D^2} \ln(\rho_1/\rho_2)$$

+

$$\frac{1}{4} \ln(\rho_1/\rho_2)$$

=

$$0$$

The scale postulate governs common realized scale.

The harmonic equation governs relational closure between the two c-grade vantages.

For the two vantages to describe the same local x-y geometry, PMM adopts the **vantage-coherence condition**:

$$x_1^D/y_1^D = x_2^D/y_2^D$$

Define the common planar readouts:

$$x^D = \sqrt{(x_1^D x_2^D)}$$

$$y^D = \sqrt{(y_1^D y_2^D)}$$

and the planar-shape coordinate:

$$a = \frac{1}{2} \ln(x^D/y^D)$$

Then:

$$x^D = L e^a$$

$$y^D = L e^{-a}$$

The variables therefore separate two different geometric roles:

r = imbalance between the c^1 and c^2 vantages

a = x-y shape within their common spatial plane

This separation prevents c-order grade from being identified with spatial axis.

7. Depth

Depth is different.

Depth is generated by the relation between the two coherent D vantages. From:

$$x^D = \sqrt{(x_1^D x_2^D)}$$

$$y^D = \sqrt{(y_1^D y_2^D)}$$

and:

$$\rho_1 = \sqrt{(x_1^D y_1^D)}$$

$$\rho_2 = \sqrt{(x_2^D y_2^D)}$$

we have:

$$x^D y^D = \rho_1 \rho_2 = L^2$$

Therefore define the internal/local depth readout:

$$z^D = L = \sqrt{(x^D y^D)}$$

Depth is therefore the common relational scale generated across the two D vantages. Neither c^1 nor c^2 is identified with a single spatial axis.

7.1 FTU Spatial Readout

Across the full $B \rightarrow D \rightarrow A$ event:

$$F_{tu}^{xy} = (\psi(x,y,z), xyz, xy)$$

B: $\psi(x,y,z)$ = potential x/y/z content before spatial realization

D: xyz = active spatial geometry

A: xy = retained post-spatial representation after active depth z ends

The B entry uses $\psi(x,y,z)$ as the spatial-potential form of the B state; the canonical state notation remains Ψ^B .

Under a pure c-grade rebalance:

$$\rho_1 \rightarrow \mu \rho_1$$

$$\rho_2 \rightarrow \rho_2 / \mu$$

L and z^D remain invariant.

At closure:

$$r \rightarrow 0$$

so:

$$\rho_1 = \rho_2 = L$$

The two c-grade vantages reach scale agreement. Closure does not require $x^D = y^D$; planar shape is carried independently by a.

8. Closure Spread

Define the closure spread across the two D-vantage scales:

$$\Sigma = \rho_1 + \rho_2 - 2z^D$$

Using:

$$\rho_1 = Le^r$$

$$\rho_2 = Le^{-r}$$

$$z^D = L$$

gives:

$$\Sigma = 2L(\cosh r - 1)$$

or:

$$\Sigma = 4L \sinh^2(r/2)$$

Therefore:

$$\Sigma \geq 0$$

and:

$$\Sigma = 0$$

$$\Leftrightarrow$$

$$r = 0$$

$$\Leftrightarrow$$

$$\rho_1 = \rho_2 = z^D = L$$

Σ measures how far the two active D vantages are from relational completion. It does not measure x-y planar anisotropy.

9. C₂ — Native D Evolution

C₂ controls what happens during D.

The primitive D evolution is written on the two vantage scales:

$$L = \sqrt{(\rho_1 \rho_2)} = \ell_E \theta^D$$

and:

$$\begin{aligned} & d^2/d\theta^{D^2} \ln(\rho_1/\rho_2) \\ & + \\ & \frac{1}{4} \ln(\rho_1/\rho_2) \\ & = \\ & 0 \end{aligned}$$

For compact analysis:

$$r = \frac{1}{2} \ln(\rho_1/\rho_2)$$

so the same system becomes:

$$L = \ell_E \theta^D$$

and:

$$r'' + \frac{1}{4}r = 0$$

PMM 3.7.1 takes the single-FTU closure interval as the explicit closure postulate:

$$\theta_{cl} = \pi$$

P5 is stated as the **minimal linear conservative closure ansatz**. Require the closure law to be linear, autonomous, second-order, undriven, undamped, and time-reversal symmetric. The minimal equation is then:

$$r'' + \kappa_{cl} r = 0$$

With the entry-symmetry condition $r'(0)=0$ and the first zero required at $\theta^D=\pi$, the coefficient is conditionally derived:

$$\kappa_{cl} = \frac{1}{4}$$

PMM adopts the harmonic class, and the numerical coefficient follows inside that class.

With the **entry-symmetry condition**:

$$r(0) = r_0$$

$$r'(0) = 0$$

where PMM adopts $r'(0)=0$ as the minimal time-symmetric entry condition,

the solution is:

$$r(\theta^D) = r_0 \cos(\theta^D/2)$$

Expansion-Contraction Readout

The existing D equations supply two derived readouts across the same angular progression:

$$L = \ell_E \theta^D$$

$$r = r_0 \cos(\theta^D/2)$$

The common realized scale L expands with θ^D , while the magnitude of the c-grade vantage imbalance contracts toward zero throughout $0 < \theta^D < \pi$ when $r_0 \neq 0$. Define the normalized readouts:

$$X^D = q = \theta^D/\pi$$

$$C^D = \theta_{\text{out}} = \cos(\theta^D/2)$$

Across the single-FTU D interval $0 \leq \theta^D \leq \pi$:

$$X^D : 0 \rightarrow 1$$

$$C^D : 1 \rightarrow 0$$

Expansion and contraction are therefore two readouts of the same D evolution. They summarize the existing D scale law and harmonic closure solution; they introduce no separate postulate, conjugacy relation, or conservation law.

So D simultaneously carries:

outwardness = common realization scale growth

inwardness = c-grade vantage closure

Interpretation:

D is outward realization plus inward completion.

C_2 is therefore:

what happens in the channel

10. Angular Closure

The single-FTU closure postulate is:

$$\theta_{cl} = \pi$$

Using the minimal harmonic closure law:

$$r(\theta^D) = r_0 \cos(\theta^D/2)$$

gives:

$$r(\pi) = 0$$

Therefore at closure:

$$r = 0$$

$$\Sigma = 0$$

$$\rho_1 = \rho_2 = z^D = L_{cl}$$

with:

$$L_{cl} = \pi \hbar c / E^D$$

The common planar readouts remain:

$$x^D = L_{cl} e^a$$

$$y^D = L_{cl} e^{-a}$$

so closure requires agreement between the two c-grade vantages rather than $x^D = y^D$.

Anti-contact is:

$$-@$$

$$\Leftrightarrow$$

$$\theta^D = \pi$$

$$\Leftrightarrow$$

$$r = 0$$

$$\Leftrightarrow$$

$$\Sigma = 0$$

The fixed π closure is a single-FTU postulate. The harmonic form is the minimal linear conservative closure ansatz; once that class is adopted, the coefficient $\frac{1}{4}$ follows from requiring the first closure at π .

11. Space Window

Normalize D progression:

$$q = \theta^D / \theta_{cl}$$

with:

$$0 \leq q \leq 1$$

Legacy phase-progress notation: Earlier PMM versions used u as a generic monotonic parameter ordering $B \rightarrow D \rightarrow A$. In the current formulation, native D progression is expressed through angular progress θ^D , with $q = \theta^D / \pi$ because $\theta_{cl} = \pi$. Where older figures or equations show $\theta_{in}(u)$, $\theta_{out}(u)$, or $\theta_{win}(u)$, u should be read as the earlier generic phase-progress parameter rather than as a separate physical variable.

Legacy window terminology: Earlier PMM versions called θ_{in} the **Birth Sigmoid** and θ_{out} the **Record Sigmoid**. These names remain valid for backward compatibility as descriptors of the opening and closing roles. The current active functions are angular half-window factors; the word “sigmoid” no longer specifies their mathematical form.

PMM adopts the **lowest-mode symmetric space-window ansatz**:

$$\theta_{in} = \sin(\pi q / 2)$$

$$\theta_{out} = \cos(\pi q / 2)$$

The window is required to satisfy:

$$\theta_{win}(0) = 0$$

$$\theta_{win}(1) = 0$$

$$\theta_{win}(q) > 0 \text{ for } 0 < q < 1$$

$$\theta_{win}(q) = \theta_{win}(1 - q)$$

$$\max \theta_{win} = 1$$

Those conditions admit multiple functions. PMM selects the smooth positive fundamental mode with no internal nodes or extra oscillations. Within that class:

$$\theta_{win}'' + \pi^2 \theta_{win} = 0$$

with the endpoint conditions above uniquely selects:

$$\theta_{win} = \sin(\pi q)$$

PMM chooses the half-window factors as the symmetric factorization:

$$\Theta_{\text{in}} = \sin(\pi q/2)$$

$$\Theta_{\text{out}} = \cos(\pi q/2)$$

Define this FTU's internal/local spatial-window readout:

$$\Theta_{\text{win}} = 2 \Theta_{\text{in}} \Theta_{\text{out}}$$

therefore:

$$\Theta_{\text{win}} = \sin(\pi q)$$

The active spatial readout is:

$$(x, y, z)$$

$$=$$

$$\Theta_{\text{win}}(x^D, y^D, z^D)$$

So:

$$@:$$

$$q = 0$$

$$\Theta_{\text{win}} = 0$$

$$D:$$

$$0 < q < 1$$

$$\Theta_{\text{win}} > 0$$

$$-@:$$

$$q = 1$$

$$\Theta_{\text{win}} = 0$$

The two D vantages reach relational completion as the active spatial window closes. Θ_{win} is a D-space manifestation/access window; it is not a measure of whether the underlying internal D geometry exists.

This separates:

internal D geometry

from:

active spatial manifestation

12. Outwardness and Inwardness

PMM phase orientation is:

B = outward possibility

D = outward realization + inward closure

A = inward persistence

Technical form:

$$dL/d\theta^D > 0$$

represents outward realization.

$$d|r|/d\theta^D < 0$$

throughout $0 < \theta^D < \pi$ when $r_0 \neq 0$ represents inward completion.

The two tendencies coexist during D.

13. Massive Sector

For a massive D realization:

$$E^D = mc^2$$

therefore:

$$\ell_E = \hbar c / E^D$$

$$= \hbar / (mc)$$

and:

$$L = (\hbar / (mc)) \theta^D$$

The intuitive D time readout is:

$$T^D = d/c$$

Massive realization accumulates proper time.

14. Null Sector

The null sector has its own native definition.

For a null carrier:

$$E\gamma = \hbar\omega$$

therefore:

$$\ell_E = \hbar c / E\gamma$$

$$= c / \omega$$

and:

$$L\gamma = (c/\omega) \theta^D$$

with:

$$d\theta^D = \omega dt$$

so:

$$dL\gamma = c dt$$

therefore:

$$dL\gamma/dt = c$$

while:

$$d\tau^D = 0$$

Light therefore progresses through D by phase progression while accumulating zero proper time.

This is the PMM meaning of:

light as stillness

The massive and null sectors share:

$$\ell_E = \hbar c / E$$

$$L = \ell_E \theta^D$$

$$x^D = L e^r$$

$$y^D = L e^{-r}$$

$$z^D = L$$

$$\Sigma = 2L(\cosh r - 1)$$

$$r \rightarrow 0 \rightarrow -@$$

15. D → A Transcription

At D closure:

$$r = 0$$

$$\Sigma = 0$$

$$L = L_{cl}$$

Define the completed D phase rate through the energy-frequency relation:

$$\omega^D = E^D / \hbar$$

PMM then adopts the D→A phase-frequency continuity postulate across −@:

$$\omega^A = \omega^D$$

therefore:

$$\omega^A = E^D / \hbar$$

and:

$$E^A = \hbar \omega^A$$

so:

$$E^A = E^D$$

Energy continuity is therefore **derived from** the D→A phase-frequency continuity postulate together with $E = \hbar \omega$; it is not an independent postulate.

The D→A phase-frequency continuity fixes the intrinsic unit-span transcription scale:

$$T^{A*} = 1 / \omega^A$$

$$= \hbar / E^D$$

The physical A record-span time is:

$$T^A = T^{A*} \Delta \theta^A$$

$$= \Delta \theta^A / \omega^A$$

The minimal A record is:

$$R^A = \{\omega^A, a_{cl}, I^A\}$$

where:

$$a_{cl} = \frac{1}{2} \ln(x_{cl}^D / y_{cl}^D)$$

$$I^A = -\ln p_j@$$

The universal closure angle $\theta_{cl} = \pi$ need not be stored event-by-event. From ω^A , PMM reconstructs $E^A = \hbar \omega^A$ and therefore $L_{cl} = (\hbar c / E^A) \theta_{cl}$. Together with the surviving planar-shape coordinate a_{cl} , the completed x-y readout is reconstructible through $x_{cl}^D = L_{cl} e^{a_{cl}}$ and $y_{cl}^D = L_{cl} e^{-a_{cl}}$. Native z depth is not written into A.

C_3 is therefore:

what persists and is transcribed

16. A Phase — Persistent Record

A is post-spatial.

It stores completed event information. Active xyz geometry belongs to D, and active z depth terminates at the $D \rightarrow A$ boundary.

Interpretation:

$$B = \text{Wave}$$

$$D = \text{Particle}$$

$$A = \text{Cube}$$

Technical interpretation:

$$\text{Wave} = \text{possibility representation}$$

$$\text{Particle} = \text{realized spatial representation}$$

$$\text{Cube} = \text{completed-event representation}$$

The A cube is an event-record representation.

Entropy belongs to the persistent A record. For the actualized channel $j@$, define:

$$I^A = -\ln p_j@$$

$$\Omega_A = e^{I^A} = 1/p_j@$$

$$S^A = k_B I^A = k_B \ln \Omega_A$$

Ω_A is the effective A-side multiplicity of the realized event. PMM therefore has the dimensionless A information span:

$$\Delta\theta^A = \mathcal{S}^A/k_B = \ln \Omega_A = -\ln p_j@$$

The associated A phase time and action readouts are:

$$T^A = [\hbar/E^A]\Delta\theta^A = \Delta\theta^A/\omega^A$$

$$S^A = E^A T^A = \hbar \Delta\theta^A$$

The intrinsic unit-span transcription scale remains $T^{A*} = 1/\omega^A$; the physical A record span need not equal one unit cycle.

Status: A information-span readout tied directly to the actualized B contact weight and carried through completed D realization.

17. Geometry and Space

PMM distinguishes geometry from space.

Geometry spans $B \rightarrow D \rightarrow A$.

But:

ordinary xyz space is active during D.

Therefore:

B = pre-spatial geometry

D = spatial geometry

A = post-spatial geometry

The FTU is geometric throughout the event.

Only D carries active xyz spatial realization.

This is the internal geometry as seen from one FTU. It is not a claim that this FTU alone constitutes complete physical space — the full geometry is relational among FTUs, per the core principle.

18. Single-FTU Vantage: Internal Geometry

The canonical single-FTU spine is:

$$B$$

$$\downarrow \Psi^B$$

$$C_1$$

$$\downarrow C_1(\Phi) \rightarrow \{\Pi_j\}$$

$$\downarrow p_j@ = \langle \Psi^B | \Pi_j | \Psi^B \rangle / \sum_k \langle \Psi^B | \Pi_k | \Psi^B \rangle$$

$$@$$

$$\downarrow P1: \lim_{\theta^D \rightarrow 0^+} \rho_1 / \rho_2 = p_1@ / p_2@$$

$$\downarrow r_0 = \frac{1}{2} \ln(p_1@ / p_2@)$$

$$\downarrow$$

$$D$$

Primitive D vantages:

$$c^1 \mapsto V_1 = (x_1^D, y_1^D)$$

$$c^2 \mapsto V_2 = (x_2^D, y_2^D)$$

$$\rho_1 = \sqrt{(x_1^D y_1^D)}$$

$$\rho_2 = \sqrt{(x_2^D y_2^D)}$$

Vantage geometry:

$$L = \sqrt{(\rho_1 \rho_2)}$$

$$r = \frac{1}{2} \ln(\rho_1 / \rho_2)$$

$$L = \ell_E \theta^D$$

$$r'' + \frac{1}{4}r = 0$$

Vantage coherence and common spatial readout:

$$x_1^D/y_1^D = x_2^D/y_2^D$$

$$x^D = \sqrt{(x_1^D x_2^D)}$$

$$y^D = \sqrt{(y_1^D y_2^D)}$$

$$a = \frac{1}{2} \ln(x^D/y^D)$$

$$z^D = L = \sqrt{(x^D y^D)}$$

$$\Sigma = \rho_1 + \rho_2 - 2z^D$$

$$\Theta_{\text{win}} = \sin(\pi q)$$

$$\text{As } \theta^D \rightarrow \pi:$$

$$r \rightarrow 0$$

$$\Sigma \rightarrow 0$$

$$\rho_1 \rightarrow \rho_2$$

$$-@$$

$$\downarrow$$

$$C_3$$

$$\downarrow \omega^A = E^D/\hbar; E^A = E^D$$

$$A$$

$$\downarrow R^A = \{\omega^A, a_{cl}, I^A\}$$

persistent record

19. Single-FTU Formal Status

PMM 3.7.1 presents the single-FTU vantage as a formal dependency structure. Every active statement in Part II is classified as a **primitive, definition/readout, postulate/ansatz, or derivation**.

19.1 Primitives

B, D, A — phase architecture

@, -@ — phase boundaries

θ^D — angular D progression

E^D — D carrier energy

c^1, c^2 — D realization grades

$V_1 = (x_1^D, y_1^D), V_2 = (x_2^D, y_2^D)$ — two D spatial vantages

These form the starting objects of the single-FTU construction.

19.2 Definitions and Readouts

$$\rho_1 = \sqrt{(x_1^D y_1^D)}$$

$$\rho_2 = \sqrt{(x_2^D y_2^D)}$$

$$\ell_E = \hbar c / E$$

$$L = \sqrt{(\rho_1 \rho_2)}$$

$$r = \frac{1}{2} \ln(\rho_1 / \rho_2)$$

$$x^D = \sqrt{(x_1^D x_2^D)}$$

$$y^D = \sqrt{(y_1^D y_2^D)}$$

$$a = \frac{1}{2} \ln(x^D / y^D)$$

$$z^D = L$$

$$\Sigma = \rho_1 + \rho_2 - 2z^D$$

$$q = \theta^D / \theta_{cl}$$

$$X^D = q = \theta^D / \pi \text{ — normalized D expansion readout}$$

$$C^D = \Theta_{out} = \cos(\theta^D / 2) \text{ — normalized D contraction readout}$$

$$\omega^D = E^D / \hbar$$

$$F_{tu}^M, F_{tu}^E, F_{tu}^T \text{ — primary FTU readouts}$$

$$F_{tu}^{xy} = (\psi(x, y, z), xyz, xy) \text{ — spatial readout}$$

$$F_{tu}^s = S^B + S^D + S^A \text{ — additive action ledger}$$

$$dS^D = \hbar d\theta^D \text{ — D angular-action normalization}$$

A information/readout definitions:

$$I^A = -\ln p_i @$$

$$\Omega_A = 1 / p_i @$$

$$\Delta\theta^A = S^A / k_B = \ln \Omega_A = I^A$$

$$S^A = \hbar \Delta\theta^A$$

These introduce notation and read out specified states. Because both $\Delta\theta^B = \arccos\sqrt{p_j@}$ and $\Delta\theta^A = -\ln p_j@$ depend on the same contact probability, the B and A spans are explicitly coupled through $p_j@$.

19.3 Explicit Single-FTU Postulates / Ansätze

P0 c-order correspondence:

$$B = (c^0)_{mp}$$

$$D = (c^1, c^2)c$$

$$A = (c^3)k_B$$

PA-B candidate B projective action span:

$$\Delta\theta^B = \arccos\sqrt{p_j@}$$

$$S^B = \hbar\Delta\theta^B$$

C₁-B candidate reciprocal closure probability:

$$p_j = |a_j|^2 / \sum_k |a_k|^2$$

C₁-C candidate contact-channel selection:

$$C_1(\Phi) \rightarrow \{\Pi_j\}$$

$$p_j@ = \langle \Psi^B | \Pi_j | \Psi^B \rangle / \sum_k \langle \Psi^B | \Pi_k | \Psi^B \rangle$$

P1 B→D ratio transfer at contact:

$$\lim_{\theta^D \rightarrow 0^+} \rho_1 / \rho_2 = p_1@ / p_2@$$

P2 D scale law:

$$L = \ell_E \theta^D$$

P3 D-vantage coherence:

$$x_1^D / y_1^D = x_2^D / y_2^D$$

the two c-grade vantages describe one common local x-y geometry

P4 angular closure:

$$\theta_{cl} = \pi$$

P5 minimal linear conservative closure class:

$$r'' + \kappa_{cl} r = 0$$

with $r'(0)=0$ as the entry-symmetry condition

and first closure at π

$$\Rightarrow \kappa_{cl} = 1/4$$

P6 lowest-mode symmetric space window:

zero endpoints, positive interior, midpoint symmetry, unit peak

plus no internal nodes / no extra oscillations

$$\Rightarrow \Theta_{win} = \sin(\pi q)$$

with $\Theta_{in} = \sin(\pi q/2)$, $\Theta_{out} = \cos(\pi q/2)$

P7 D→A phase-frequency continuity:

$$\omega^A = \omega^D$$

These are the irreducible assumptions and conditional-selection principles that govern the single-FTU vantage. The D angular-action relation is classified as a definition/readout in §19.2 rather than as an additional postulate. C₁-B and C₁-C are candidate entry rules rather than derived laws. P5 and P6 are selected within their adopted minimal admissible classes.

19.4 Derived Results

From the definitions and postulates above:

$$r_0 = 1/2 \ln(p_1@/p_2@)$$

$$\rho_1 = L e^r$$

$$\rho_2 = L e^{-r}$$

$$x^D = L e^a$$

$$y^D = L e^{-a}$$

$$x^D y^D = L^2$$

$$z^D = L = \sqrt{(x^D y^D)}$$

$$\Sigma = 2L(\cosh r - 1)$$

$$= 4L \sinh^2(r/2)$$

$$\Sigma \geq 0$$

$$\Sigma = 0 \Leftrightarrow r = 0 \Leftrightarrow \rho_1 = \rho_2 = z^D$$

$$\kappa_{\text{cl}} = 1/4$$

$$r(\theta^D) = r_0 \cos(\theta^D/2)$$

$$X^D = q = \theta^D/\pi : 0 \rightarrow 1$$

$$C^D = \theta_{\text{out}} = \cos(\theta^D/2) : 1 \rightarrow 0$$

$$L_{\text{cl}} = \pi \hbar c / E^D$$

$$\Theta_{\text{win}} = \sin(\pi q)$$

$$E^A = E^D$$

$$I^A = -\ln p_i @$$

$$\Omega_A = 1/p_i @$$

$$\Delta\theta^A = I^A = \ln \Omega_A$$

$$\mathcal{S}^A = k_B I^A$$

$$S^A = \hbar I^A$$

$$T^{A*} = \hbar / E^D$$

$$S^D(\theta^D) = \hbar \theta^D$$

$$S^D, \text{cl} = \pi \hbar$$

$$F_{\text{tu}}^{\text{s}, \text{cl}} = \hbar (\Delta\theta^B + \pi + \Delta\theta^A)$$

The coefficient $1/4$ follows from P4 plus the harmonic form in P5. Depth follows from the coherent two-vantage construction. Energy continuity follows from P7 plus the energy-frequency relation. The Expansion-Contraction Readout follows from P2 together with the derived harmonic closure solution.

The standard identity $\ell_p^2 = \hbar G_N / c^3$ appears in the c-order profile and in the local c^3 discussion. The candidate gravitational-crease condition $A_{cl} = A_G$, when applied to the current π closure geometry, yields $E_{cl}^D = \pi m_p c^2$; this remains a candidate limiting condition pending a native C_3 selection law.

19.5 Massive / Null Consistency

The single-FTU geometry includes both carrier sectors:

massive:

$$E^D = mc^2$$

$$\ell_E = \hbar / (mc)$$

proper time accumulates

null:

$$E\gamma = \hbar\omega$$

$$\ell_E = c/\omega$$

$$d\theta^D = \omega dt$$

$$dL\gamma/dt = c$$

$$d\tau^D = 0$$

$$dS^D = \hbar d\theta^D$$

The null sector has its own native definition. The massive and null sectors share the same L, r, a, z^D, Σ geometry, angular-action readout, and closure condition.

19.6 Minimality / Open Single-FTU Work

The current model structurally specifies the single-FTU architecture. Remaining work focuses on reducing its postulates:

OPEN derive or reduce P0, the c-order correspondence, from deeper PMM structure if possible

OPEN derive or reduce the C_1 contact-channel selection rule from deeper PMM structure

OPEN test the candidate reciprocal closure probability and closure-basis invariance

OPEN test or derive the candidate B projective span $\Delta\theta^B = \arccos\sqrt{p_j@}$

OPEN derive or reduce P1 if a deeper $B \rightarrow D$ principle is found

OPEN derive or reduce P2 if angular time uniquely fixes the D scale law

OPEN derive P3 if D-vantage coherence follows from a deeper PMM symmetry

OPEN derive P4 from a deeper closure principle

OPEN test whether the minimal linear conservative class in P5 is itself forced

OPEN test whether the lowest-mode condition in P6 is itself forced

OPEN derive or reduce P7 if transcription continuity follows from C_3 itself

19.7 Canonical Single-FTU Spine

B potential Ψ^B

↓

C_1 -C: C_1 selects admissible contact channels $\{\Pi_i\}$

↓

contact weights $p_i@$ from projection of Ψ^B

↓

@

↓

P1: $p_1@/p_2@ \rightarrow \lim_{\theta^D \rightarrow 0^+} \rho_1/\rho_2 = e^{2r_0}$

↓

$c^1 \mapsto V_1 = (x_1^D, y_1^D)$

$c^2 \mapsto V_2 = (x_2^D, y_2^D)$

D action definition: $dS^D = \hbar d\theta^D$

↓

$\rho_1, \rho_2 \rightarrow L, r$

P2: $L = \ell_E \theta^D$

derived D pair: $X^D = q$ rises $0 \rightarrow 1$ while $C^D = \Theta_{out}$ falls $1 \rightarrow 0$

P3: vantage coherence $\rightarrow$ common x^D, y^D and $z^D=L$

P4/P5: π closure + minimal linear conservative class $\Rightarrow \kappa_{cl} = 1/4$

P6: lowest-mode window $\Rightarrow \Theta_{win} = \sin(\pi q)$

↓

$r \rightarrow 0, \Sigma \rightarrow 0, \rho_1 \rightarrow \rho_2$

↓

-@

↓

P7: $\omega^A = \omega^D$

↓

A persistent record

This is the canonical single-FTU dependency chain for PMM 3.7.1.

19.8 Recovery Methodology / Firewall

PMM keeps native construction separate from recovery. A recovery begins with PMM primitives and derived relations, applies a stated limit or bridge, and then compares the result with established physics.

A successful recovery match does not promote the assumptions, variables, or equations used for comparison into PMM first principles. Recovery confirms compatibility within the stated limit; it does not redefine the native foundation.

19.9 Candidate Test / Falsification Discipline

Every candidate relation states its dependencies, test conditions, and failure condition before adoption. A failed test rejects or revises the candidate. A passed test establishes consistency with the stated test; it does not by itself convert the candidate into a first principle or derivation.

Promotion requires a PMM-native derivation, or a new independently testable consequence that survives its stated falsification test.

PART III — MULTI-FTU VANTAGE

20. Multi-FTU Principle

The single-FTU vantage describes the internal geometry of one complete event.

The multi-FTU vantage describes the relational geometry among many FTUs.

The two are separate sectors of one theory:

single FTU

→ local event geometry

multi FTU

→ shared relational geometry

They couple through the event-record cycle:

single FTU

→ completed A record

→ multi-FTU field

→ future C_1 / C_2 conditions

→ later single FTUs

Thus PMM is one recursive theory expressed through two vantages.

21. Primitive Relational Geometry

The core relational quantity is local interaction depth:

δ = local interaction depth

For a pair of FTUs:

$$\delta_{ij} = \delta_{ji}$$

$$\delta_{ii} = 0$$

$$\delta_{ij} > 0 \text{ for } i \neq j$$

The relational distance matrix is:

$$M\delta = \{\delta_{ij}\}$$

No global direction vector is required as a primitive. Direction can be reconstructed after the distance network is embedded.

The distinction is:

$$z^D = \text{internal single-FTU D depth}$$

$$\delta_{ij} = \text{inter-FTU relational depth}$$

δ_{ij} is an **independent primitive of the multi-FTU sector**. The direct bridge between single-FTU z^D and inter-FTU δ is an open derivation target.

The two vantages couple through completed FTU outputs and the shared history/source sector; the direct $z^D \leftrightarrow \delta$ bridge is open.

22. Shared Space from FTU Relations

One FTU gives an internal event view.

Two FTUs give a separation.

Three compatible FTUs give the first closed relational loop.

Four suitably independent FTUs give relational three-dimensional volume.

1 FTU → internal event view

2 FTUs → relational separation

3 FTUs → relational area / loop

4 FTUs → relational 3D volume

Pairwise δ_{ij} relations assemble shared physical space.

The single-FTU architecture is ordinal and angular, while many overlapping FTU relations build shared physical geometry. PMM treats an effectively smooth macroscopic spacetime as a candidate recovery: an appropriate many-FTU coarse-graining limit would make the relational network approach a continuum description at observable scales. The specific limit and bridge remain OPEN.

For four FTUs, the six consistent pair distances define a tetrahedral relational cell. Such cells assemble larger spatial geometry.

23. Discrete Curvature

When relational tetrahedra assemble around an edge, curvature can be represented by the deficit angle:

$$\Delta_e = 2\pi - \sum_t \theta_{te}$$

where θ_{te} is the dihedral angle contributed by tetrahedron t around edge e .

Then:

$$\Delta_e = 0 \rightarrow \text{locally flat assembly}$$

$$\Delta_e \neq 0 \rightarrow \text{relational curvature}$$

Curvature emerges from the assembled δ geometry.

23.1 Recovery Bridge

The discrete relational network supplies the native multi-FTU geometry. Standard differential geometry then provides a recovery bridge from that network to continuum curvature:

$$\delta\text{-network} \rightarrow \text{local frame} \rightarrow \text{connection} \rightarrow \text{curvature} \rightarrow R$$

This bridge belongs to recovery. Standard frame, connection, and curvature tensors act downstream of the PMM relational structure rather than as PMM first principles.

24. Relational Event Lattice

A multi-FTU state at relational slice n is:

$$\mathcal{M}^n = \{\delta_{ij}^n, \Phi_i^n, T_i^{D,n}\}$$

At the next slice:

$$\mathcal{M}^{n+1} = \{\delta_{ij}^{n+1}, \Phi_i^{n+1}, T_i^{D,n+1}\}$$

The network therefore contains two kinds of relations:

spatial edge:

$$F_i^n \leftrightarrow F_j^n$$

$$\text{length} = \delta_{ij}^n$$

temporal edge:

$$F_i^n \rightarrow F_i^{n+1}$$

$$\text{local D-time interval} = \Delta T_i^{D,n}$$

Repeated relational slices produce an effective event geometry:

$$\mathcal{M}^0 \rightarrow \mathcal{M}^1 \rightarrow \mathcal{M}^2 \rightarrow \dots$$

This is the PMM route from local FTU geometry to emergent spacetime.

25. A-Side Record and Recursive History

A completed FTU contributes a persistent A record.

The history update is:

$$\Phi_i^{n+1} = \text{Append}(\Phi_i^n, R_i^{A,n})$$

where Append means that the completed record $R_i^{A,n}$ is added to the persistent history carried by Φ ; it is a record-accumulation operation, not ordinary scalar addition.

For the gravity sector, Φ_s denotes a scalar readout of the accumulated record Φ . Ordinary addition applies to Φ_s only in the reduced regime of §30.1; the general Append rule is not assumed to be additive.

The core constraint set already couples Φ back into later events:

$$\begin{aligned} \Theta_{\text{con}} = \{ \\ & C_1[@,\Phi], \\ & C_2[\Phi,\delta,\Sigma], \\ & C_3[\Phi,\Sigma] \\ & \} \end{aligned}$$

So the larger PMM cycle is:

$$B \rightarrow D \rightarrow A \rightarrow \Phi \rightarrow \text{future } B/D$$

Interpretation:

$C_1 = \Phi$ conditions future contact admissibility

$C_2 = \Phi$ conditions realized D propagation

$C_3 =$ completed D writes persistent A content

Φ is therefore the memory channel by which completed geometry conditions future realization.

26. A-Side Energy and Stress Record

A completed FTU carries an energetic readout:

$$E^A = \hbar\omega^A = E^D$$

with mass-equivalent readout:

$$m^A = E^A/c^2$$

The minimal mass-density contribution over relational volume V^* is:

$$\rho^A = E^A / (c^2 V^*)$$

A complete gravitational source can also carry the completed D event's response to deformation. The revised two-vantage geometry separates common scale, planar shape, and c-grade imbalance before any stress law is proposed.

26.1 Revised D Deformation Coordinates

The common spatial readouts are:

$$x^D = L e^a$$

$$y^D = L e^{-a}$$

$$z^D = L$$

Therefore the logarithmic spatial strains are:

$$\varepsilon_x = d \ln x^D = d \ln L + da$$

$$\varepsilon_y = d \ln y^D = d \ln L - da$$

$$\varepsilon_z = d \ln z^D = d \ln L$$

so:

$$\varepsilon_z = \frac{1}{2}(\varepsilon_x + \varepsilon_y)$$

The two c-grade vantage scales are:

$$\rho_1 = L e^r$$

$$\rho_2 = L e^{-r}$$

with differential responses:

$$d \ln \rho_1 = d \ln L + dr$$

$$d \ln \rho_2 = d \ln L - dr$$

Thus a and r carry different deformation information:

$$a = \text{planar x-y shape}$$

$$r = c^1\text{-}c^2 \text{ vantage imbalance}$$

At anti-contact, $r \rightarrow 0$; planar shape a need not vanish.

26.2 Native D Energy Functional — Open

No active candidate D interaction-energy law is adopted in this release. The native interaction energy remains an open derivation target. Its most general current dependence can be written schematically as:

$$E^D = E^D(L, a, r, \theta^D, \text{carrier data}, \dots)$$

Any future candidate must be specified independently and then tested against the existing PMM geometry. It must admit the massive and null sectors and support acceptable stress behavior without defining an auxiliary variable from the same equation being tested.

26.3 Stress Readout

Directional stress remains a downstream readout of an independently supplied D interaction energy functional:

$$p_u = -(1/V) \partial E^D / \partial \varepsilon_u$$

with $u \in \{x, y, z\}$ and spatial strain channels ε_x , ε_y , and ε_z .

The c-grade imbalance r can also produce an internal vantage-response channel once a native E^D is known:

$$I_r^D = -(1/V) \partial E^D / \partial r$$

I_r^D is an internal D response coordinate, not automatically a standard spatial pressure. A bridge to ordinary stress components requires derivation.

At closure, C_3 transcribes the completed energy-and-response content into the A record. Let ρ_E denote the local energy density. In a local rest-frame recovery representation:

$$\mathcal{R}_{\mu\nu}^A \rightarrow \text{diag}(\rho_E, p_1, p_2, p_3)$$

and many completed FTUs can build an effective stress-energy source:

$$T_{\mu\nu} = (1/V^*) \sum \text{completed } \mathcal{R}_{\mu\nu}^A$$

26.4 Status

Current status:

two-vantage D deformation geometry

→ active

$$\varepsilon_z = \frac{1}{2}(\varepsilon_x + \varepsilon_y)$$

→ derived from $x^D = Le^a$, $y^D = Le^{-a}$, $z^D = L$

native E^D interaction functional

→ OPEN

spatial stress and internal vantage response

→ downstream of a future independently specified E^D functional

The principal remaining derivation target in this sector is therefore a native D interaction energy functional compatible with the revised two-vantage geometry.

27. Multi-FTU Gravity Lineage

The gravity sector begins at the multi-FTU vantage and uses relational clock variables rather than c-order labels as direct dynamical inputs.

Its canonical lineage is:

completed FTU

→ persistent record R^A

→ accumulated history Φ

→ C_2 conditions later D realization

→ local D angular-rate response κ

→ relational clock potential $\varphi = -\ln \kappa$

→ gravity as a relational D-clock gradient

→ Record Influence Continuity

→ Clock Flux

The single-FTU c-order architecture defines what an FTU is. The multi-FTU gravity architecture describes how completed FTUs influence later D-phase timing.

Every step is tagged below as one of:

CORE — already present in the PMM architecture

DEFINITION — introduces notation and readouts

DERIVED — follows from prior stated assumptions

POSTULATE — new physical content not forced by prior mathematics

CANDIDATE — proposed relation not yet derived

RECOVERY — comparison or normalization against established physics

OPEN — active derivation target

Compound labels combine these statuses when a statement has more than one role.

This status discipline is part of the canonical theory. Recovery status leaves upstream inputs classified according to their native PMM role.

28. Native D Clock Response

28.1 Core inputs

CORE

PMM states:

Time is angular.

$$F_{tu}^T = ([\hbar/(m_p c^2)] \Delta\theta^B, d/c, \Delta\theta^A/\omega^A)$$

$$F_{tu}^G = \nabla\varphi$$

$$g = c^2 F_{tu}^G$$

$$C_2 = C_2[\Phi, \delta, \Sigma]$$

Thus gravity is associated with normalized relational variation in D-phase timing, while accumulated completed history Φ conditions later D realization through C_2 .

The gravitational variable uses a dimensionless comparison of D angular rates across FTUs with different intrinsic D scales. This relational multi-FTU construction uses normalized D-clock behavior as its direct dynamical input.

28.2 Dimensionless angular-rate response

DEFINITION

For FTU i , compare realized D angular progression with its intrinsic neutral progression over the same relational update:

$$\kappa_i = \Delta\theta_i^D / \Delta\theta_{i,0}^D$$

where:

$$\kappa_i = 1 \rightarrow \text{neutral D progression}$$

$$\kappa_i < 1 \rightarrow \text{slowed D progression}$$

$$\kappa_i > 1 \rightarrow \text{accelerated D progression}$$

For a common angular increment $\Delta\theta^D$, let T_i^D be the D-clock interval required to accumulate that increment and $T_{i,0}^D$ its neutral reference interval. Then:

$$\kappa_i = (\Delta\theta^D/T_i^D)/(\Delta\theta^D/T_{i,0}^D) = T_{i,0}^D/T_i^D$$

so:

$$T^D_i/T^D_{i,0} = 1/\kappa_i$$

For a massive realization the same elapsed interval may be read through $T^D = d/c$. The gravity sector uses the ratio to the neutral D-clock interval, which is dimensionless and remains well defined across different intrinsic D scales.

Because κ is defined from angular progression, the null sector uses the same clock framework:

null carrier:

$$d\tau^D = 0$$

but

$$\Delta\theta^D \neq 0$$

The angular-rate definition keeps the gravitational clock variable defined in the null sector.

29. Relational Clock Potential and Gravity

29.1 Common-rate invariance

A gravitational relation is invariant when every local clock response is rescaled by the same positive factor:

$$\kappa_i \rightarrow S\kappa_i$$

$$\kappa_j \rightarrow S\kappa_j$$

Edge observables depend on relative clock behavior.

DEFINITION

Introduce the dimensionless relational clock potential:

$$\varphi_i = -\ln \kappa_i$$

Using the D-clock relation above:

$$\varphi_i = \ln(T^D_i/T^D_{i,0})$$

Then a common rescaling shifts every φ by the same constant and therefore cancels from all differences.

For relational edge $i \leftrightarrow j$:

$$\Gamma_{ij} = (\varphi_j - \varphi_i)/\delta_{ij}$$

Γ_{ij} is the dimensionless D-clock gradient per relational length.

29.2 Gravitational acceleration readout

DEFINITION / CORE-MOTIVATED READOUT

The D crease c converts the inverse-length clock gradient to acceleration:

$$g_{ij} = c^2 \Gamma_{ij}$$

Continuum form:

$$g = c^2 \nabla \varphi$$

Equivalently, since $\varphi = -\ln \kappa$:

$$g = -c^2 \nabla \ln \kappa$$

Using $\varphi = \ln(T^D/T^D_0)$:

$$g = c^2 \nabla \ln(T^D/T^D_0)$$

For a spatially uniform neutral reference T^D_0 :

$$g = (c^2/T^D) \nabla T^D$$

Thus g points toward increasing T^D , the locally slower D-clock side, consistent with the attractive sign convention below.

The normalized FTU gravity readout is therefore:

$$F_{tu}^G = \nabla \varphi = \nabla \ln(T^D/T^D_0)$$

with:

$$g = c^2 F_{tu}^G$$

Gravity is therefore the relational gradient of normalized D-clock response. This keeps the human-facing D-time interpretation while making the dimensions explicit.

29.3 Sign convention

A positive completed source slows local D progression:

source present

$$\rightarrow \kappa < 1$$

$$\rightarrow \varphi > 0$$

For an isolated attractive source, φ decreases outward. Therefore:

$\nabla \varphi$ points inward

and:

$$g = c^2 \nabla \varphi$$

has the correct attractive orientation.

30. How Φ Conditions the D Clock

The multi-FTU source-response chain is:

$$\text{completed A record} \rightarrow \Phi \rightarrow C_2 \rightarrow \text{later D realization}$$

The full function relating accumulated history to local clock response remains open.

The general native form is therefore:

$$\kappa_i = K(\Phi_i, \text{local } \delta\text{-relations}, \Sigma_i, \dots)$$

OPEN

The complete function K is not yet derived.

30.1 Reduced additive-history candidate

If, in a reduced regime with the other arguments held fixed:

$$\Phi_{s,12} = \Phi_{s,1} + \Phi_{s,2}$$

and independent history contributions compose multiplicatively in their effect on D rate:

$$\kappa(\Phi_{s,1} + \Phi_{s,2}) = \kappa(\Phi_{s,1})\kappa(\Phi_{s,2})$$

then continuity and positivity give:

$$\kappa(\Phi_s) = e^{-\alpha \Phi_s}$$

for some normalization α .

Hence:

$$\varphi = -\ln \kappa = \alpha \Phi_s$$

DERIVED CONDITIONAL ON THE MULTIPLICATIVE-RESPONSE ASSUMPTION

The canonical clock field is $\varphi = -\ln \kappa$. Global proportionality between φ and Φ_s is an open derivation.

31. Record Influence Continuity

C_3 already provides persistence:

$$C_3 = \text{what persists and is transcribed into } \Phi$$

and the multi-FTU history update accumulates completed records:

$$\Phi_i^{n+1} = \text{Append}(\Phi_i^n, R_i^{A^n})$$

Record Influence Continuity supplies the spatial flux principle.

31.1 Gravity-sector postulate

POSTULATE — Record Influence Continuity

A completed A record conditions later D realization through the relational network. Away from completed-record sources, its clock influence is locally conserved.

It contains two claims:

locality:

influence propagates through FTU relations

continuity:

source-free relational regions conserve clock influence

This postulate defines the relational continuity principle upstream of the continuum field equation.

32. Clock Flux

32.1 Source-free consequence

Given Record Influence Continuity, a source-free relational region has zero net production of the clock-gradient field.

DERIVED FROM THE POSTULATE

In continuum notation:

$$\nabla \cdot (\nabla \varphi) = 0$$

or:

$$\nabla^2 \varphi = 0$$

in a source-free region after the continuum limit is valid.

The discrete form is primary. The continuum equation is the candidate large-network representation, pending derivation of the many-FTU coarse-graining limit.

32.2 Enclosed-source form

For a closed relational region V^R , continuity requires the outward boundary flux to equal the negative enclosed completed-record source multiplied by one coupling normalization:

$$\oint \partial V^R \nabla \varphi \cdot dA = -\lambda_G Q^A(V^R)$$

where:

$$Q^A(V^R) = \text{total completed gravitational record-source inside } V^R$$

and:

$$[\lambda_G] = \text{length} / \text{mass}$$

if Q^A is represented as a mass-equivalent source.

The negative sign is fixed by the convention that positive source gives $\varphi > 0$ near the source while outward flux is negative.

This is the **Clock Flux Law**.

Status:

Record Influence Continuity — POSTULATE

Clock Flux Law — DERIVED from continuity + locality + additive enclosed source

coupling normalization λ_G — OPEN until fixed by recovery or deeper derivation

32.3 Minimal A-side source candidate

The single-FTU readouts already provide:

$$E^A = E^D$$

$$m^A = E^A / c^2$$

The minimal scalar source candidate is therefore:

$$Q^A = \sum \text{completed } m^A$$

$$= \sum \text{completed } E^A / c^2$$

CANDIDATE

This is the lowest-order source identification. The full relativistic source can also include pressure, anisotropic stress, and additional completed-record structure.

33. Discrete δ -Network Clock Flux

The multi-FTU theory is fundamentally relational, so the discrete law precedes the continuum law.

For each relational edge $i \leftrightarrow j$:

$$\Gamma_{ij} = (\varphi_j - \varphi_i) / \delta_{ij}$$

Let A^*_{ij} denote the dual relational face area associated with boundary edge $i \leftrightarrow j$ in the assembled tetrahedral network.

For relational region V^R :

$$\sum_{i \leftrightarrow j \in \partial V^R} A^*_{ij} \Gamma_{ij}$$

$$= -\lambda_G Q^A(V^R)$$

This is the discrete Clock Flux equation.

For an interior node i, the corresponding discrete divergence form is:

$$\begin{aligned} \sum_j [A^*_{ij}/\delta_{ij}] (\varphi_j - \varphi_i) \\ = -\lambda_G q^A_i \end{aligned}$$

where q^A_i is the completed record-source assigned to the node or its dual cell.

Status:

δ_{ij} — existing multi-FTU primitive

A^*_{ij} — derived from the chosen relational cell complex

φ — relational D-clock potential

source-free discrete balance — derived from Record Influence Continuity

source coupling λ_G — OPEN

No global spatial direction field is required.

34. Consistency Checks

The following checks test the gravity chain before downstream GR recovery.

34.1 Dimensional consistency

φ	dimensionless
$\nabla\varphi$	L^{-1}
$c^2\nabla\varphi$	$L T^{-2}$

so:

$$g = c^2\nabla\varphi$$

has acceleration dimensions.

For the flux law:

$$\oint \nabla\varphi \cdot dA \quad L$$

therefore a mass-equivalent source requires:

$$[\lambda_G] = L/M$$

and:

$$[G_N/c^2] = L/M$$

provides the accepted gravitational crease scale.

34.2 Common-rate invariance

Under:

$$\kappa_i \rightarrow S\kappa_i$$

we have:

$$\varphi_i \rightarrow \varphi_i - \ln s$$

so edge differences and gravity are unchanged.

34.3 Massive-sector kinematic consistency

Massive FTUs have nonzero D progression and therefore admit κ directly.

The massive sector is kinematically consistent.

34.4 Null-sector kinematic consistency

Null carriers satisfy:

$$d\tau^D = 0$$

and carry angular D progression:

$$d\theta^D \neq 0$$

Because κ is an angular-rate ratio, the gravitational clock variable is defined for the null sector.

34.5 A-record recursion

The source lineage is:

D completion

$$\rightarrow C_3$$

$$\rightarrow R^A$$

$$\rightarrow \Phi$$

$$\rightarrow \text{later } C_2$$

$$\rightarrow \kappa$$

so gravity is downstream of completed record persistence.

34.6 Attractive sign convention

Positive source:

$$\kappa < 1$$

→ $\varphi > 0$ near source

→ φ decreases outward

→ $\nabla\varphi$ points inward

→ $g = c^2\nabla\varphi$ points inward

The sign is internally consistent.

34.7 Newtonian normalization — RECOVERY

RECOVERY

If the minimal source is:

$$Q^A = M$$

then matching the weak-field Newtonian Gauss law fixes:

$$\lambda_G = 4\pi G_N/c^2$$

so:

$$\oint \nabla\varphi \cdot dA = -(4\pi G_N/c^2) M$$

and:

$$\oint g \cdot dA = -4\pi G_N M$$

with local continuum form:

$$\nabla^2\varphi = -(4\pi G_N/c^2) \rho$$

and:

$$\nabla \cdot g = -4\pi G_N \rho$$

Matching the weak-field Newtonian limit fixes the coupling by recovery. G_N enters as the accepted gravitational crease, and $Q^A = M$ supplies the minimal scalar recovery source.

34.8 Foundational target

The foundational gravity target is:

Derive the full completed-record source Q^A from PMM's event architecture and determine the native status of λ_G relative to the accepted gravitational crease G_N .

35. Single-FTU / Multi-FTU Coupling

PMM presents the two sectors separately and couples them dynamically.

Single FTU supplies:

$B \rightarrow D \rightarrow A$ event geometry

D angular progression

E^A

R^A

C_3 persistence

Multi-FTU supplies:

δ network

accumulated history Φ

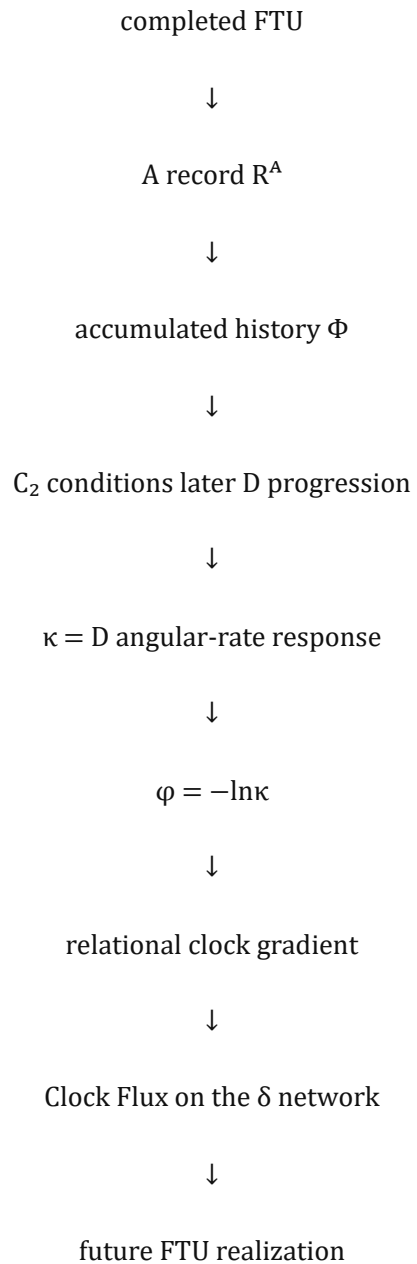
relational D-clock response κ

clock potential φ

clock-gradient field

future C_1/C_2 conditions

The feedback cycle is:



This defines one recursive theory with two vantages:

$$\text{single FTU} = \text{creases} + \text{c-order} + \text{action}$$

$$\text{multi-FTU} = \Phi + \text{timing response} + \text{gravity}$$

36. Theory Status

Single-FTU vantage — foundation

Core structure: creases + c-order + action

B/D/A event architecture
 @ and -@
 c-order mapping
 two x-y D vantages
 relational depth from coherent vantages
 L/r/a decomposition
 closure spread Σ
 $\theta_{cl} = \pi$
 minimal harmonic closure
 space window
 massive sector
 null sector
 A transcription
 energy continuity
 FTU readout family
 phase-action ruler with $dS^D = \hbar d\theta^D$ and variable B/A spans
 candidate B span $\Delta\theta^B = \arccos\sqrt{p_j@}$ from the C_1 -selected contact channel
 A information span $\Delta\theta^A = I^A = -\ln p_j@ = \ln \Omega_A$ with $\Omega_A = 1/p_j@$
 completed action profile $\Delta\theta^B \rightarrow \pi \rightarrow \pi \rightarrow \Delta\theta^A$ across the c-order readout
 outwardness / inwardness

Single-FTU refinement:

derive or reject native C_3 selection of $A_{cl} = A_G$ at the gravitational-crease limiting boundary

Multi-FTU vantage — relational scaffold

Core structure: Φ + timing response + gravity

CORE

δ_{ij} as independent relational depth

$A \rightarrow \Phi \rightarrow \text{future-event recursion}$

$$F_{tu}^G = \nabla\varphi = \nabla\ln(T^D/T^D_0)$$

$$g = c^2 F_{tu}^G$$

$$C_2[\Phi, \delta, \Sigma]$$

DEFINITION / CORE-MOTIVATED READOUT

$\kappa = D$ angular-rate response

$$\varphi = -\ln \kappa$$

$$\Gamma_{ij} = \Delta\varphi/\delta_{ij}$$

$$g = c^2 \nabla \varphi$$

POSTULATE

Record Influence Continuity

DERIVED FROM THE POSTULATE

source-free clock-flux conservation

closed-region Clock Flux form

discrete δ -network flux equation

CANDIDATE

$$Q^A = \sum E^A/c^2 \text{ as minimal scalar source}$$

$$\kappa(\Phi_s) = e^{-\alpha\Phi_s} \text{ in the reduced additive/multiplicative regime}$$

OPEN

native D interaction energy functional of §26

RECOVERY

$$\lambda_G = 4\pi G_N/c^2 \text{ from the Newtonian weak-field normalization}$$

Canonical gravity foundation boundary

The canonical gravity foundation consists of the relational δ geometry, completed-record history Φ , D-phase clock response, and the normalized D-clock-gradient readout $F_{tu}^G = \nabla\varphi$. Exploratory constructions from earlier releases that are not defined in this document are not inputs to the canonical gravity foundation.

Open derivations

native C_3 selection or rejection of $A_{cl} = A_G$ at the gravitational-crease limiting boundary

full native function $\kappa = K(\Phi, \delta, \Sigma, \dots)$

full completed-record gravitational source Q^A

native value of λ_G , if derivable beyond accepting G_N

native D interaction-energy and stress derivation

A-side stress-energy transcription

direct $z^D \leftrightarrow \delta$ bridge, if one exists

fully discrete δ -network evolution in time

many-FTU coarse-graining / continuum limit from the discrete relational network
covariant continuum limit
independent predictions beyond recovery

The gravity derivation concentrates on the completed-record source, native clock response, stress transcription, and continuum limit.

37. Research Continuation

Current research priorities are maintained in [ROADMAP.md](#).

Revision history is maintained in [CHANGELOG.md](#).

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